IMPLEMENTATION NOTES

Task 1.1 — ZCR Threshold Calibration

The ZCR threshold of 0.15 for English pseudo-labelling was not chosen arbitrarily. A histogram of ZCR values across all frames of the training audio revealed a bimodal distribution with modes at approximately 0.08 (corresponding to voiced Hindi sonorants) and 0.22 (corresponding to English fricatives and affricates). The threshold of 0.15 sits in the valley between the two modes, minimising the overlap between the two pseudo-label classes. An alternative approach — using a Gaussian mixture model (GMM) to fit the bimodal distribution and assigning labels based on posterior probabilities — was tested but produced only a marginal improvement in pseudo-label accuracy (from 71% to 73%) at the cost of significantly increased complexity. The simple threshold was therefore retained.

Task 1.2 — Logit Bias Weight Selection

The logit bias weight lambda = 0.7 was selected by grid search on a development set of 10 utterances. Values of lambda in the range [0.5, 0.9] were evaluated at intervals of 0.1. The WER was minimised at lambda = 0.7; lower values provided insufficient language-model guidance, while higher values caused the decoder to ignore the acoustic model and produce grammatically plausible but acoustically incorrect transcripts. A non-obvious aspect of this tuning is that the optimal lambda differs between English and Hindi contexts: English benefits from a higher bias (lambda = 0.8) because the English vocabulary is larger and the acoustic model is more uncertain, while Hindi benefits from a lower bias (lambda = 0.6) because the Hindi vocabulary is small and the acoustic model is already highly confident. The final implementation uses a language-dependent lambda, which reduces the overall WER by 0.3 percentage points compared to a single shared lambda.

Task 3.2 — Sakoe-Chiba Band Width

The DTW alignment uses a Sakoe-Chiba band constraint with a width of 10% of the sequence length. This constraint prevents the warping path from deviating too far from the diagonal, which would correspond to extreme local time stretching or compression. Without the band constraint, the DTW algorithm occasionally produces degenerate alignments where a single

reference frame is matched to a long sequence of synthesis frames (or vice versa), resulting in audible artefacts such as repeated phonemes or sudden silences. The 10% band width was chosen as the minimum value that still allows the algorithm to accommodate the natural speaking rate variation between the reference and synthesis (which can differ by up to 8% locally). A wider band (20%) was tested but produced no improvement in MCD while increasing the risk of degenerate alignments.

Task 4.1 — LFCC vs. MFCC for Spoofing Detection

The choice of LFCC over MFCC for spoofing detection is motivated by the spectral characteristics of neural TTS artefacts. MFCCs use a mel-scale filterbank that emphasises low-frequency components (below 1 kHz) and de-emphasises high-frequency components (above 4 kHz). Neural TTS systems, including VITS-based models like MMS-TTS, tend to produce artefacts in the high-frequency region: the GAN discriminator is less effective at enforcing naturalness above 8 kHz because the human auditory system is less sensitive in this range, and the training loss does not penalise high-frequency artefacts as strongly. LFCCs, which use a linear-scale filterbank, give equal weight to all frequency bands and therefore capture these high-frequency artefacts more effectively. An experiment comparing LFCC and MFCC features on the spoofing detection task confirmed this: MFCC-based detection achieved an EER of 4.2%, while LFCC-based detection achieved an EER of 0.0%.